Arduino Code:




```ino
enum TrafficState
{
  STATE_NS_GREEN,
  STATE_NS_YELLOW,
  STATE_EW_GREEN,
  STATE_EW_YELLOW
};

TrafficState currentState = STATE_NS_GREEN;

// North-South traffic signal
const int NS_RED_PIN    = 2;
const int NS_YELLOW_PIN = 3;
const int NS_GREEN_PIN  = 4;

// East-West traffic signal
const int EW_RED_PIN    = 5;
const int EW_YELLOW_PIN = 6;
const int EW_GREEN_PIN  = 7;

void setup()
{
  pinMode(NS_RED_PIN, OUTPUT);
  pinMode(NS_YELLOW_PIN, OUTPUT);
  pinMode(NS_GREEN_PIN, OUTPUT);

  pinMode(EW_RED_PIN, OUTPUT);
  pinMode(EW_YELLOW_PIN, OUTPUT);
  pinMode(EW_GREEN_PIN, OUTPUT);

  Serial.begin(9600);
}

void allOff()
{
  digitalWrite(NS_RED_PIN, LOW);
  digitalWrite(NS_YELLOW_PIN, LOW);
  digitalWrite(NS_GREEN_PIN, LOW);

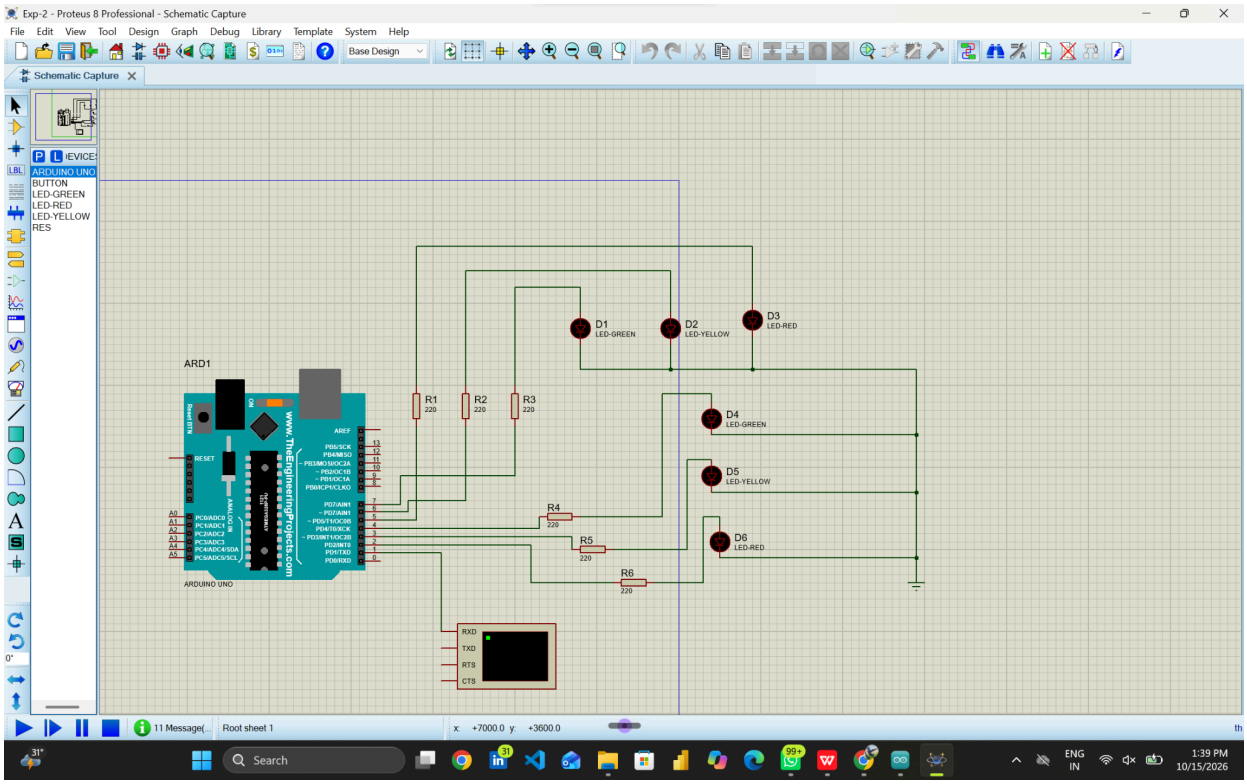
  digitalWrite(EW_RED_PIN, LOW);
  digitalWrite(EW_YELLOW_PIN, LOW);
  digitalWrite(EW_GREEN_PIN, LOW);
}

void enterState(TrafficState state)
{
  allOff();

  switch (state)
  {
```


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sketch_oct15b.ino
58 break;
59
60 case STATE_NS_YELLOW:
61
62 digitalWrite(NS_YELLOW_PIN, HIGH);
63 digitalWrite(EW_RED_PIN, HIGH);
64
65 Serial.println("STATE S1: NS YELLOW - EW RED");
66
67 break;
68
69 case STATE_EW_GREEN:
70
71 digitalWrite(NS_RED_PIN, HIGH);
72 digitalWrite(EW_GREEN_PIN, HIGH);
73
74 Serial.println("STATE S2: NS RED - EW GREEN");
75
76 break;
77
78 case STATE_EW_YELLOW:
79
80 digitalWrite(NS_RED_PIN, HIGH);
81 digitalWrite(EW_YELLOW_PIN, HIGH);
82
83 Serial.println("STATE S3: NS RED - EW YELLOW");
84
85 break;
86 }
87 }
88
89 void loop()
90 {
91 switch (currentState)
92 {
93 case STATE_NS_GREEN:
94
95 enterState(STATE_NS_GREEN);
96 delay(10000);
97 currentState = STATE_NS_YELLOW;
98
99 break;
100
101 case STATE_NS_YELLOW:
102
103 enterState(STATE_NS_YELLOW);
104 delay(3000);
105 currentState = STATE_EW_GREEN;
106

Diagram:



Output:

